JOSE FIGUEROA

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EDUCATION

University of California, Riverside Expected June 2027
B.S. Chemical Engineering
- **Relevant coursework:** Engineering Modeling & Analysis, Process Control, Chemical Engineering Laboratory, Fluid Dynamics, Reaction Engineering, Separations, Transport Phenomena.

College of the Desert May 2020
A.S. Chemistry — Palm Desert, CA

RESEARCH EXPERIENCE

Undergraduate Researcher 2026 – Present
Petters Aerosol Laboratory, CE-CERT — University of California, Riverside
- Run size-selected cloud condensation nuclei (CCN) activation experiments on a CCN-100 counter paired with a dual differential mobility analyzer and condensation particle counter.
- Developed an eddy-diffusivity wall-loss correction for SMPS chamber size distributions, applied per size bin so reported distributions describe the suspended aerosol rather than what survives to the inlet.
- Lead developer of the laboratory's Python and Streamlit analysis application: automatic instrument file detection, clock-offset correction between instruments, CCN and DMA/CPC run merging, a searchable results library, and live LabJack/Raspberry Pi streaming.

Research Intern Summer 2026
Mentoring Summer Research Internship Program (MSRIP) — UC Riverside
- **Awarded Best Research Presentation** at the close of the program; served as head of safety and procedures for the research team.

TECHNICAL PROJECTS

- **Aerosol Instrument Dashboard** — Python, Streamlit, pandas, LabJack. In daily use in the Petters lab; removes the manual reconciliation step where measurement errors were entering the record.
- **Lung Deposition Explorer** — maps a measured particle size distribution to deposited fraction in the head, tracheobronchial and alveolar regions, reading a distribution as exposure rather than counts.
- **ChemE Excel Toolkit & Panel** — Excel add-in and task pane providing NIST Antoine coefficients, vapor-liquid equilibrium and ternary liquid-liquid extraction lookups inside any workbook.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, Streamlit), MATLAB, C++, VBA
Simulation: Aspen Plus V14
Aerosol instrumentation: SMPS, DMA, CPC, CCN-100, LabJack, Raspberry Pi
Analytical: UV-Vis, IR, GC, TLC; spectral interpretation and technical reporting
Other: Advanced Excel, laboratory safety protocols and SOPs for hazardous materials

WORK EXPERIENCE

Catering & Service June 2025 – Present
Foundation 10 — Palm Springs, CA
- Operate estate and large-scale events, including off-site catering preparation and food truck service; identify and rework inefficient workflows to keep a team moving during service.

Certified Pharmacy Technician 2015 – 2025
CVS Pharmacy — Indio, CA
- Prepared COVID-19 prescriptions, tests and vaccines alongside the pharmacist while handling high prescription volume; flagged potential drug interactions and allergies across all ages and conditions.
- Verified insurance benefits, read formularies and coordinated exceptions with insurers; transmitted daily bulk orders and managed drug inventory control.

CERTIFICATION

Certified Pharmacy Technician (CPhT) — License TCH 48772